

A guide of R programs for “Adaptive spectral analysis for nonstationary functional time series”

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Abstract

This supplementary file provides a guide to the R implementing the proposed estimator in the paper “Adaptive spectral analysis for nonstationary functional time series”.

To use these functions, users should download and unzip `Rprogs_FPCA_AFSE.zip` to a suitable directory. The file is available at https://github.com/zhangshibin-repository/BayesInfe_FunctionSpec. It contains the main R scripts `algorithm1.R`, `algorithm2.R`, and `algorithm3.R`, together with several supporting scripts. After downloading, the required scripts can be loaded using the following commands:

```
source(algorithm1.R); source(model.R); source(hmc_gibbs.R); source(birth.R)
source(death.R); source(within.R); source(algorithm3.R); source(algorithm2.R)
```

The proposed estimator (Algorithm 2) is implemented by the function `samc` in `algorithm2.R`. This function mainly calls Algorithm 1 (for FPC dimension selection) and Algorithm 3 (for SAMC sampling), and then completes the spectral density operator reconstruction.

Usage:

```
samc(X, M=8, t_min=120, J=10, delta=0.05, fd_nbasis=21, sigma_a=100, sigma_b=100, c_tau2=100,
     a1=1, a2=2, a_tau=0.3, a_vartheta=0.3, laplace_maxit=1000, laplace_ridge=1e-12,
     seed=20260910, rho=0.8, relocation_radius=2, epsilon0=0.02, trajectory_length=0.4, T0=10,
     n_iter=10000, burn_in=2000, thin_sample=10, thin_time=10, omega=NULL)
```

Arguments:

<code>X</code>	Observed observed discrete functional time series as a $T \times N$ matrix, where T is the number of equally spaced grid points on $[0, 1]$, and N is the sample size.
<code>delta</code>	Accuracy level δ in Algorithm 1.
<code>fd_nbasis</code>	Number of basis functions used in the functional representation.
<code>M</code>	Maximum number of segments.
<code>J</code>	J is the J in equation (11) and (12).
<code>t_min</code>	$t_{\min}$ is the $t_{\min}$ used in Section 3.1.
<code>sigma_a, sigma_b</code>	σ_a and σ_b are the σ_a and σ_b in equation (14), respectively.
<code>c_tau2</code>	c_{τ^2} is the c_{τ^2} in equation (15).

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a1, a2	a1 and a2 are the a_1 and a_2 in equation (16), respectively.
a_tau, a_vartheta	a_tau and a_vartheta are the a_τ and a_θ used in the birth move, respectively.
rho	rho is the ϱ used in the within move.
relocation_radius	relocation_radius is the r_ξ used in the within move.
epsilon0	Baseline stepsize ϵ_0 in the HMC.
trajectory_length	Nominal trajectory length ℓ in the HMC.
T0	T_0 is the T_0 in equation (20).
laplace_maxit	Maximum number of optimization iterations in the Laplace approximations.
laplace_ridge	Hessian ridge constant in the Laplace approximations.
seed	Pre-specified random seed.
n_iter	Total number of SAMC iterations.
burn_in	Number of initial iterations discarded.
thin_sample	Thinning interval for retaining sampler states after burn-in.
thin_time	Spacing between time indices at which the spectral density kernels are reconstructed.
omega	Given frequency vector at which the spectral density kernels are reconstructed.

Values:

p	Number of FPCs.
J	J is the J in equation (11) and (12).
N, T	Sample size and number of grid points per curve, respectively.
draws	List of retained sampler states. Each state contains: 1. b: Iteration index. 2. m: Number of segments. 3. xi: Vector of segment boundaries ξ_m. 4. segments: List of the m segments. The lth element contains: (a) beta: Vector of spectral coefficients β_l. (b) tau2: Vector of smoothing variances τ_l^2. (c) vartheta: Vector of multiplicative gamma variables ϑ_l. (d) eigenfunctions: $T \times p$ matrix of the $\nu^{(l,m)}$.
time, omega	Time indices and frequencies at which the spectral density kernels are reconstructed.
kernel_mean	Complex $T \times T \times n_t \times n_\omega$ array representing the estimated spectral density operators, where n_t and n_ω are the lengths of time and omega, respectively.